

AquaRider: A Constant-Buoyancy Underwater Vessel Using Constant Buoyancy And Variable Effective Weight Control and Thrusters.

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Abstract

Conventional underwater vessels rely primarily on ballast-based buoyancy control, which involves large changes in displacement and results in slow vertical response, structural complexity, and operational limitations involving limited supply for compressed air. This paper presents **AquaRider**, a novel amphibious vessel concept that maintains *constant buoyancy* while achieving vertical and lateral maneuvering through **controlled variation of effective weight**, flexible bottom-mounted tanks, and pump-based thrusters. The system eliminates traditional propellers, reduces surface wake, and improves maneuverability and stealth. A working physical model was developed and experimentally demonstrated using low-cost components to validate the feasibility of the concept.



Keywords

Submersible vessel, constant buoyancy, variable effective weight, flexible tanks, pump thrusters, underwater maneuvering.

1. Introduction

Underwater and submersible vessels conventionally employ rigid ballast tanks to control buoyancy. While effective, these systems introduce several drawbacks, including slow depth control, increased

structural stress, and limited availability of compressed air. Additionally, propeller-based propulsion generates visible surface wake and noise, reducing stealth.

The **AquaRider** concept was developed to address these limitations by decoupling *buoyancy* from *depth control*. Instead of varying displacement, the system maintains constant buoyancy and controls motion by dynamically altering the **effective weight** of the vessel. This paper outlines the working principle, mechanical layout, and experimental validation of the AquaRider model.

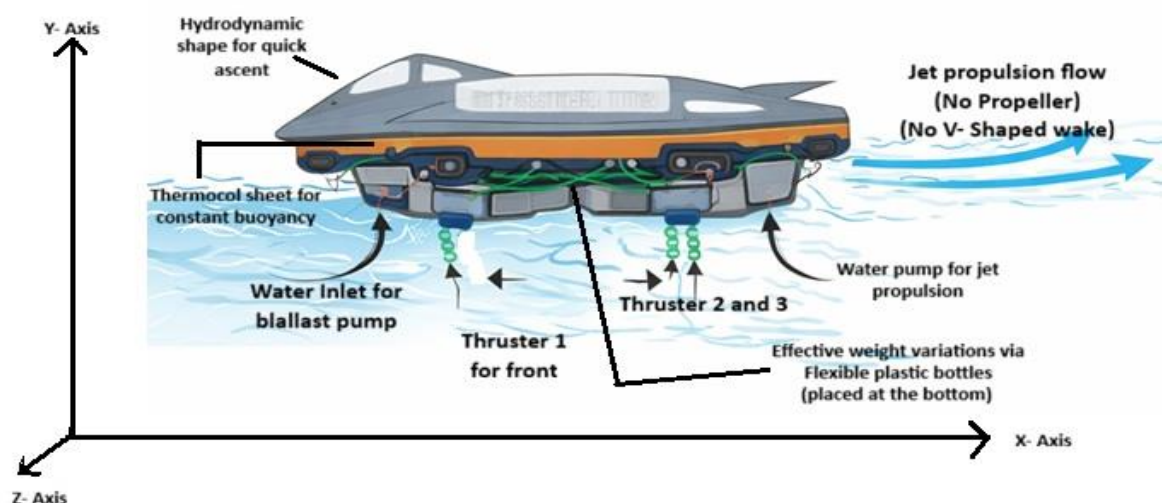
2. Design Philosophy

The design of AquaRider is based on three fundamental principles:

- 1. Constant Buoyancy Maintenance**
Buoyancy is kept constant throughout operation using a lightweight core material, eliminating the need for large ballast exchanges.
- 2. Effective Weight Modulation**
Vertical motion is achieved by changing the vessel's effective weight relative to buoyant force, rather than altering displacement.
- 3. Pump-Based Jet Propulsion**
All propulsion and maneuvering forces are generated using pump-driven water jets instead of propellers, reducing wake and mechanical complexity.

3. Structural Configuration

The vessel consists of three vertically arranged functional zones:



3.1 Buoyancy Core

The central section of the vessel contains a **thermocool (expanded polystyrene) buoyancy core**, which provides a fixed buoyant force. This ensures that the vessel's buoyancy remains constant irrespective of depth or operating mode.

3.2 Flexible Effective-Weight Tanks

At the **bottom-most section** of the vessel, multiple **flexible tanks** are installed. These tanks were fabricated using **sealed, squeezable plastic bottles with a single opening**. Their primary function is not buoyancy, but **effective weight control**. Moreover these tanks are protected with a base metal sheet.

When water is pumped into these flexible tanks, the vessel's effective weight increases. When water is extracted, the effective weight decreases. Since the tanks are flexible, **ambient water pressure at greater depths assists in squeezing the tanks**, naturally aiding the expulsion of water and improving system efficiency.

3.3 Hull Geometry

The hull is designed with a **flat, oval-shaped hydrodynamic profile** complemented by fin-like extensions. This geometry reduces drag, minimizes stress concentrations, and offers greater usable internal volume compared to conventional tear-drop submarine hulls.

4. Vertical Motion Control Mechanism

Vertical motion is achieved through **variable effective weight control**:

- Two **desert cooler pumps connected in series** are used:
 - One pump fills the flexible tanks with water.
 - The second pump extracts water from the tanks.
- As water enters the tanks, the vessel's weight exceeds buoyant force, causing descent.
- As water is expelled, the effective weight drops below buoyant force, causing ascent.

This method enables **faster and smoother depth control** compared to traditional ballast systems, without altering the vessel's overall displacement.

5. Thruster and Propulsion System

5.1 Thruster Configuration

AquaRider employs **three pump-based thrusters**, all implemented using desert cooler pumps (separate from the ballast pumps):

- **Two rear thrusters** positioned at the left and right corners of the stern.
- **One forward thruster** positioned at the bow.

This configuration enables forward motion, directional control, and lateral movement along the Z-Axis.

5.2 Thruster Orientation and Hydrodynamic Advantage

The thrusters are oriented such that:

- **Inlets face forward**, directly into the oncoming water flow.
- **Outlets face downward**, producing thrust along the vertical and lateral axes.

As the vessel moves forward, **water drag at the inlet increases inlet pressure**, thereby increasing the effective discharge pressure of the pumps. This passive pressure augmentation improves thrust efficiency without additional power input.

5.3 Lateral and Axis Control

By selectively operating the thrusters, the vessel can maneuver along its **lateral (Z) axis**, as well as maintain stability and directional control during submerged operation.

5.4 Jet Propulsion

The system of Jet propulsion involves a set of Three another pumps which are different from the thruster pumps and ballast pumps.

- These pumps are of higher capacity .
- One pump is placed at the stern of the vessel , and two are placed on the wings.
- All the three pumps together, gives the vessel a forward push.
- Maneuvering can be achieved by smartly switching between these pumps.

6. Experimental Model and Validation

A physical working model of AquaRider was constructed using:

- Thermocol sheets for constant buoyancy
- Sealed plastic bottles as flexible tanks
- Desert cooler pumps for both ballast control and propulsion

The model successfully demonstrated:

- Controlled ascent and descent without changing buoyancy
- Reduced surface disturbance during motion
- Stable maneuvering in shallow water conditions

These experiments validated the feasibility of the proposed design.



7. Advantages Over Conventional Systems

- Eliminates rigid ballast tanks
- Faster depth response
- Reduced surface wake and noise
- Low-cost, scalable, and modular design

8. Potential Applications

- Shallow-water surveillance
 - Environmental monitoring
 - Educational and experimental underwater platforms
 - Stealth marine research vehicles
 - Defence purposes
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9. Conclusion

AquaRider demonstrates that constant buoyancy combined with variable effective weight control offers a viable alternative to conventional underwater vessel designs. The successful development and testing of a functional model highlight the potential for further research, scaling, and real-world application of this concept in advanced marine systems.

10. Future Work

Future development may include:

- Closed-loop depth control
- CFD-based hydrodynamic optimization
- Advanced materials for flexible tanks
- Autonomous navigation and sensor integration